Module 2

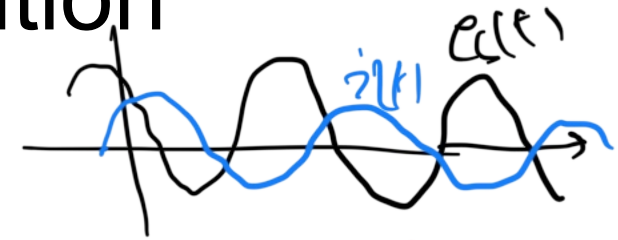
State-Variable-Based Modeling and Simulation

Most Important Topics and Concepts

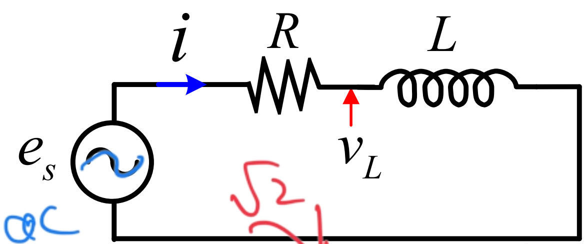
❑ Principles of:

- State-space modeling of electric circuits
- Integration methods and ODE solvers
- Numerical stability and stiffness
- Time step size control
- Multi-step methods

Electric Circuits Solution



❖ Phasor Analysis:



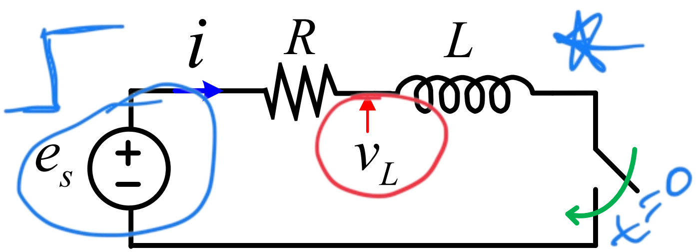
$\tilde{E}_s = |E_s| \angle \theta_s$

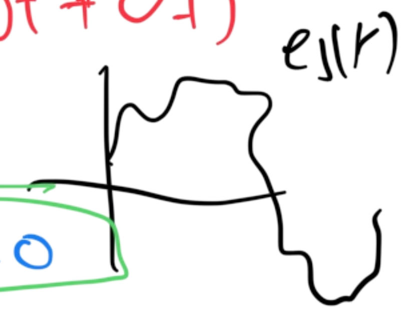
$\tilde{I}_s = \frac{\tilde{E}_s}{R+jL\omega} = |I| \angle \theta_I$

$e_s(t) = \sqrt{2}|E_s|\cos(\omega t+\theta_s)$

$i(t) = \sqrt{2}|I|\cos(\omega t+\theta_I)$

❖ Transient Analysis:





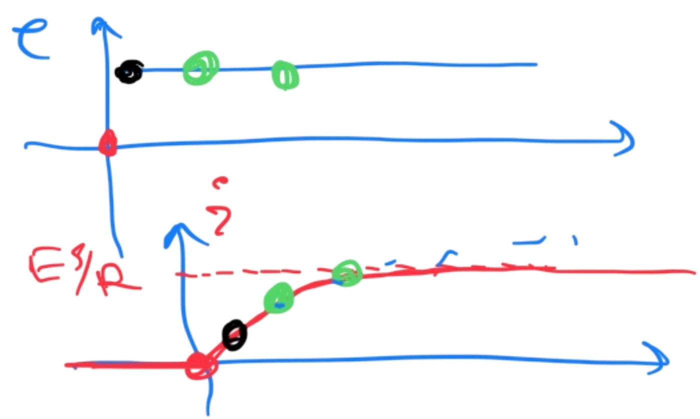
❖ Analytical Solution:

$-e_s(t) + Ri(t) + L\frac{di(t)}{dt} = 0$

$i(t) = \frac{E_s}{R}\left[1-e^{-t/\tau}\right] \quad \tau = L/R$

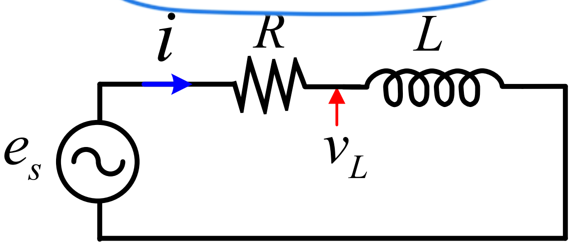
❖ Numerical Solution: (step-by-step)

t	$e_s(t)$	v_L	i
0	0	0	0
$\Delta t = 50\mu s$	100	100	✓
$2\Delta t$	100	x	α



Electric Circuits Solution

❖ **ODE vs. DAE:**



$$\frac{di}{dt} = \frac{1}{L}\left(e_s(t) - R\,i(t)\right)$$

$$-e_s(t) + v_R + v_L = 0$$

$$v_R = R\,i$$

$$L\,\frac{di}{dt} = v_L$$

$$\dot{x}(t) = \frac{dx}{dt}$$

➢ **Ordinary Differential Equations (ODEs):** $\dot{\mathbf{x}}(t) = \mathbf{f}\left(\mathbf{x}(t),\ \mathbf{u}(t),\ t\right)$

✓ All equations are differential in nature (i.e., involve derivatives of states)

➢ **Differential-Algebraic Equations (DAEs):** $\mathbf{F}\left(\dot{\mathbf{x}}(t),\ \mathbf{x}(t),\ \mathbf{z}(t),\ \mathbf{u}(t),\ t\right) = 0$

✓ Some equations do not have derivatives (i.e., are algebraic constraints)

✓ **ODEs:** Suitable for control systems and small circuits with well defined states (requires selecting independent states and eliminating algebraic constraints

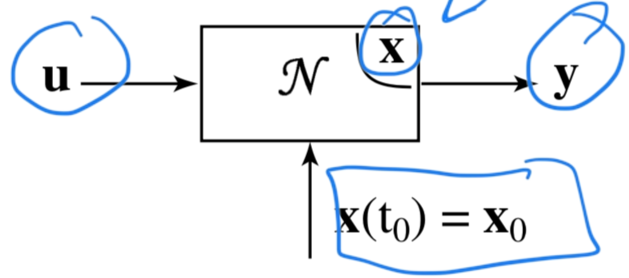
✓ **DAEs:** Scales more easily and systematically to larger networks but is harder to solve due to handling of constraints

State-Variable-Based Modeling

- ➢ Many physical systems (electrical, mechanical, thermal, etc.) are described by **differential equations**

- ✓ State-space modeling provides a **systematic way** to represent these equations in a compact vector form

❖ State-Space Form:



- ✓ A **Linear Time-Invariant (LTI)** system can be expressed as:

$$\begin{cases} \dfrac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \end{cases}$$

$\mathbf{x}$: vector of states

$\mathbf{u}$: vector of input

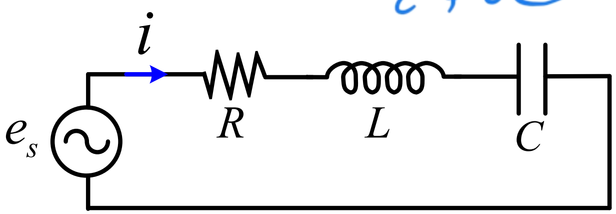
$\mathbf{y}$: vector of outputs

- ✓ **State variables** are the minimum set of variables describing the system's memory (e.g., capacitor voltages, inductor currents, etc.)

State-Variable-Based Modeling

❖ **Example 1:** Express the equations of the circuits below in state-space form:

i, v_C



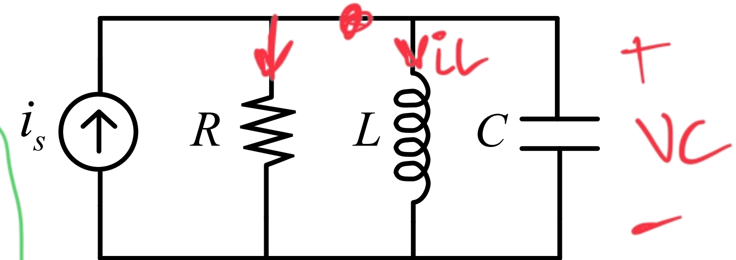
KVL:

$$e_s(t) = Ri + L\frac{di}{dt} + v_C \qquad X = \begin{bmatrix} v_C \\ i \end{bmatrix}$$

$$i = C\frac{dv_C}{dt}$$

$$\begin{cases} \dfrac{dv_C}{dt} = \dfrac{1}{C}\, i \\[2mm] \dfrac{di}{dt} = \dfrac{1}{L}\left(-v_C - Ri + e_s\right) \end{cases}$$

$$\underbrace{\frac{d}{dt}\begin{bmatrix} v_C \\ i \end{bmatrix}}_{dX/dt} = \underbrace{\begin{bmatrix} 0 & 1/C \\ -\frac{1}{L} & -\frac{R}{L} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} v_C \\ i \end{bmatrix}}_{X} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix}}_{B} \underbrace{e_s}_{u}$$



$$\begin{cases} i_s = \dfrac{v_C}{R} + i_L + C\dfrac{dv_C}{dt} \\[2mm] L\, di/dt = v_C \end{cases}$$

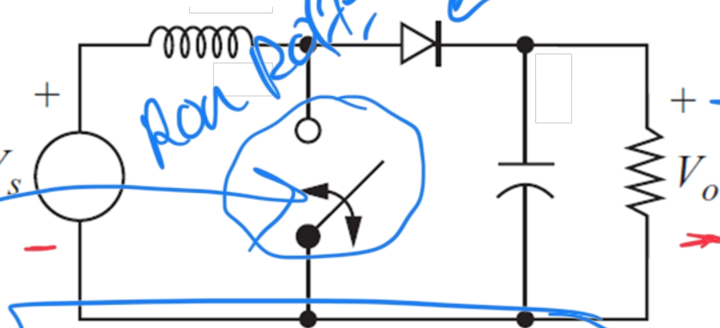
$$\begin{cases} \dfrac{dv_C}{dt} = \dfrac{1}{C}\left(-\dfrac{v_C}{R} - i_L + i_s\right) \\[2mm] di/dt = 1/L\, v_C \end{cases}$$

$$A = \begin{bmatrix} -\frac{1}{RC} & -\frac{1}{C} \\ 1/L & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1/C \\ 0 \end{bmatrix}$$

State-Variable-Based Modeling

❖ **Example 2:** Express the equations of the converter below in state-space form:



binary model of switch

$$\bar{v}_o = \frac{1}{1-D}\bar{v}_s$$

$$i_c = C\frac{dv_c}{dt} = -\frac{v_c}{R_o} + i_D$$

$$v_s = L\,di_L/dt + R_D i_D + v_c$$

$$R_s i_L - R_s i_D = R_D i_D + v_c$$

$$\boxed{R_s(i_L - i_D) = R_D i_D + v_c}$$

$$i_D = \frac{R_s}{R_s+R_D} i_L - \frac{1}{R_s+R_D} v_c$$

$$C\frac{dv_c}{dt} = -\frac{v_c}{R_o} - \frac{1}{R_s+R_D} v_c + \frac{R_s}{R_s+R_D} i_L$$

$$L\,di_L/dt = -v_c - \frac{R_D R_s}{R_s+R_D} i_L - \frac{R_D}{R_s+R_D} v_c + v_s$$

$\boxed{R_s(i_L - i_D)}$

$$\frac{d}{dt}\begin{bmatrix} i_L \\ v_C \end{bmatrix} = \overbrace{\begin{bmatrix} \frac{-R_S R_D}{L(R_S+R_D)} & \frac{-R_S}{L(R_S+R_D)} \\ \frac{R_S}{C(R_S+R_D)} & -\frac{1}{C}\left(\frac{1}{R_D}+\frac{1}{R_S+R_D}\right) \end{bmatrix}}^{A} \begin{bmatrix} i_L \\ v_C \end{bmatrix}$$

$$+ \underbrace{\begin{bmatrix} 1/L \\ 0 \end{bmatrix}}_{B} \underbrace{J_S}_{u}$$

$$X = \begin{bmatrix} i_L \\ v_C \end{bmatrix}$$

$$\frac{dX}{dt} = AX + B \cdot u$$

Transfer Function in Laplace Domain

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}$$

- ✓ Linear systems can be transformed into the **Laplace domain**

$$\boxed{\frac{d}{dt} \equiv s}$$

$$sX$$

$$\begin{cases} s \cdot X = A \cdot X + Bu \\ Y = CX + Du \end{cases}$$

- ✓ We can obtain the input-output transfer function:

$$\mathbf{H}(s) = \frac{\mathbf{y}(s)}{\mathbf{u}(s)}$$

$$sX - AX = Bu \Rightarrow (s \cdot I - A)X = Bu$$

$$X = (sI - A)^{-1} Bu$$

$$Y_{(s)} = C(sI - A)^{-1}Bu + Du \Rightarrow \left[C(sI-A)^{-1}B + D\right]u(s)$$

$$\frac{Y(s)}{u(s)} = H(s) \begin{bmatrix} H_{11}(s) & H_{12}(s) \\ H_{21}(s) & H_{22}(s) \end{bmatrix}$$

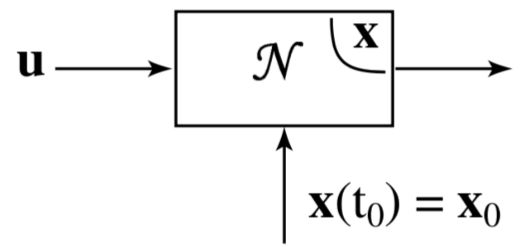
$$\mathbf{H}(s) = \mathbf{C}[s\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} + \mathbf{D}$$

State-Variable-Based Modeling

➢ Many real systems are **nonlinear** and **cannot** be expressed in a matrix form

✓ We can write the equations in a general form that can also be applicable to nonlinear systems

❑ General State-Space Form:



$$\begin{cases} \dfrac{d\mathbf{x}(t)}{dt} = \mathbf{f}\big(\underbrace{\mathbf{x}(t),\ \mathbf{u}(t),\ t}\big) \\ \mathbf{y}(t) = \mathbf{g}\big(\underbrace{\mathbf{x}(t),\ \mathbf{u}(t),\ t}\big) \end{cases}$$

$\mathbf{f},\ \mathbf{g}$: nonlinear function

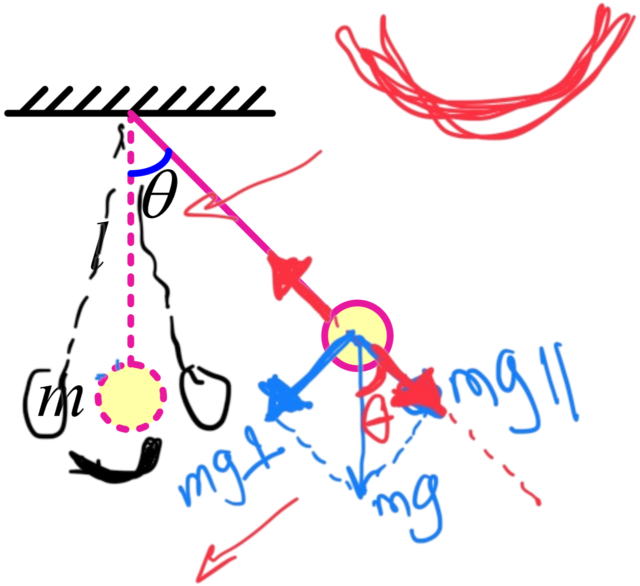
❖ Examples of Real Nonlinear Systems:

Generator with saturation

State-Variable-Based Modeling

$F = ma$

❖ **Example 3:** Express the equations of the pendulum below in state-space form:



$mg_\perp = mg \cdot \sin\theta$

$J = ml^2$

$T = \underline{mg_\perp} \times l$

$T = -J \cdot \alpha$

$(\cancel{m} g \sin\theta) \times \cancel{l} = -\cancel{m} l^{\cancel{2}} \alpha$

T: accelerating torque

$\alpha = \dfrac{d^2\theta}{dt^2}$

$$g \sin\theta = -l \frac{d^2\theta}{dt^2} \rightarrow \boxed{\frac{d^2\theta}{dt^2} = \frac{-g}{l} \sin\theta}$$

$$\begin{cases} x_1 = \theta \\ x_2 = \dot\theta \end{cases} \rightarrow \begin{cases} \dfrac{dx_1}{dt} = \dot\theta = x_2 \\ \dfrac{dx_2}{dt} = \ddot\theta = -g/l \sin(x_1) \end{cases}$$

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \cancel{\quad} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

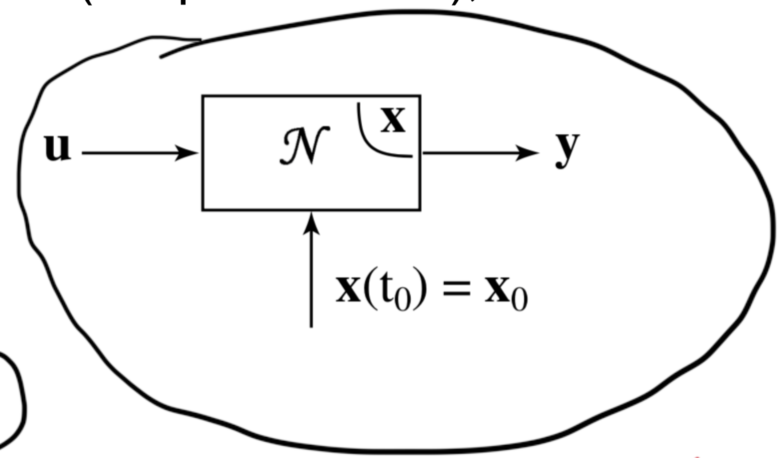
State-Space Models and Linearization

- ➢ Nonlinear systems are **hard to analyze** directly but can be converted into an **LTI** system around an operating point

- ✓ Then we can obtain transfer functions for controller tuning, small-signal stability analysis, predict transient behavior (via poles/zeros), etc.

❑ General State-Space Form:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \quad , \quad \mathbf{y} = \mathbf{g}(\mathbf{x}, \mathbf{u}, t)$$



- ➢ Assume operating around $\mathbf{x}_{op}$, $\mathbf{u}_{op}$

- ➢ Apply perturbation to the states and inputs

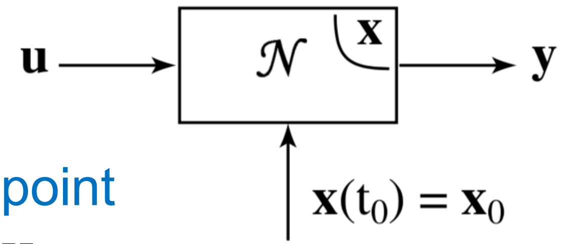
$$\begin{cases} \mathbf{x} = \mathbf{x}_{op} + \Delta\mathbf{x} \\ \mathbf{u} = \mathbf{u}_{op} + \Delta\mathbf{u} \end{cases} \Longrightarrow$$

- ✓ observe the changes in the dynamics and outputs

$$\begin{cases} \dfrac{d\Delta\mathbf{x}}{dt} = \mathbf{A}\Delta\mathbf{x} + \mathbf{B}\Delta\mathbf{u} \\ \Delta\mathbf{y} = \mathbf{C}\Delta\mathbf{x} + \mathbf{D}\Delta\mathbf{u} \end{cases}$$

State-Space Models and Linearization

❖ Linearization of Nonlinear State-Space System:



$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \quad , \quad \mathbf{y} = \mathbf{g}(\mathbf{x}, \mathbf{u}, t)$$

operating point

$\mathbf{x}_{op} \ , \ \mathbf{u}_{op}$

$$\begin{cases} \mathbf{x} = \mathbf{x}_{op} + \Delta\mathbf{x} \\ \mathbf{u} = \mathbf{u}_{op} + \Delta\mathbf{u} \end{cases} \Rightarrow \begin{cases} \dfrac{d\mathbf{x}}{dt} = ? \\ \mathbf{y} = ? \end{cases} \Rightarrow$$

Use Taylor Series!

$$\begin{cases} \dfrac{d\Delta x}{dt} = A \cdot \Delta x + B \cdot \Delta u \\ \Delta y = C \cdot \Delta x + D \cdot \Delta u \end{cases}$$

$$\underbrace{\frac{d}{dt}(x_{op} + \Delta x)} = f(\underbrace{x_{op} + \Delta x}, \underbrace{u_{op} + \Delta u}, t)$$

$$\cancel{\frac{d}{dt} x_{op}} + \frac{d}{dt}\Delta x = \cancel{f(x_{op}, u_{op}, t)} + \left.\frac{\partial f}{\partial x}\right|_{op} \Delta x + \left.\frac{\partial f}{\partial u}\right|_{op} \Delta u$$

$$y = \cancel{y_{op}} + \Delta y = \cancel{g(x_{op}, u_{op}, t)} + \left.\frac{\partial g}{\partial x}\right|_{op} \Delta x + \left.\frac{\partial g}{\partial u}\right|_{op} \Delta u$$

State-Variable-Based Modeling

State-Space of LTI System

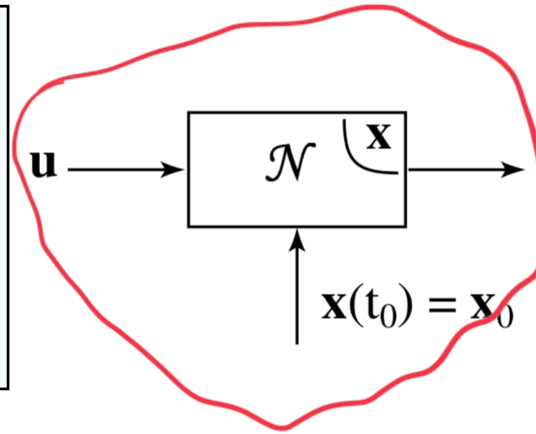
$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \boxed{\mathbf{D}\mathbf{u}} \end{cases}$$

Direct Feedthrough

General State-Space Form

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x},\ \mathbf{u},\ t) \\ \mathbf{y} = \mathbf{g}(\mathbf{x},\ \mathbf{u},\ t) \end{cases}$$

*



❑ Strictly Proper System:

When the output is not directly related to input (i.e., **no algebraic feedthrough**)

linear system: $D = 0$ nonlinear system: $y = g(x, t)$

y_1

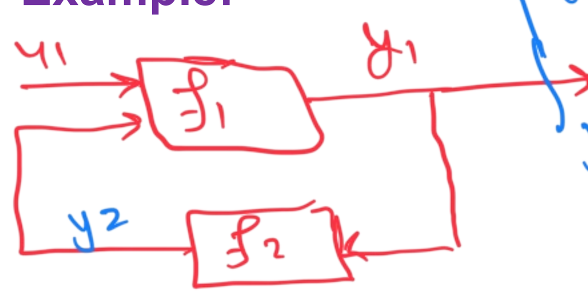
❑ Algebraic Loop:

When a variable is defined implicitly by itself through algebraic (non-differential) relations, i.e., $z = g(\mathbf{x},\ z,\ t)$

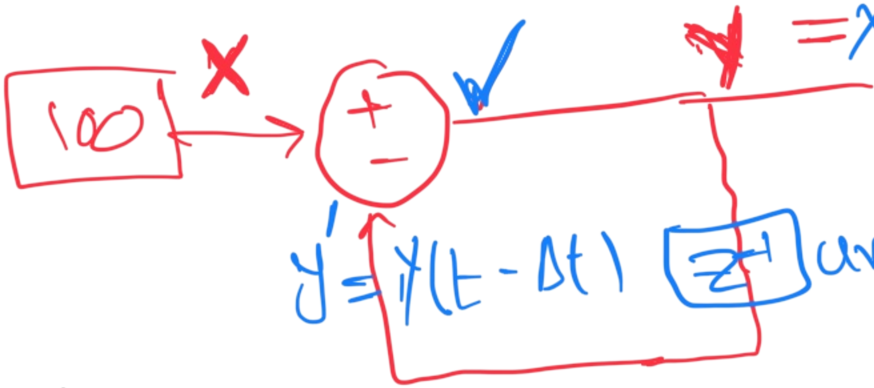
you must solve the algebraic equation first (often iteratively)

can slow down simulations!

❖ **Example:**



$$\begin{cases} y_1 = f_1(u_1, y_2^*) \\ y_2 = f_2(y_1^*) \end{cases}$$



$y = x - y(t - \Delta t)$

$y' = y(t - \Delta t)$

$y = x - y$

$\boxed{y = \frac{1}{2} x}$

t	x	y'	y
0	100	50	50
Δt	100	50	50

$\boxed{y = x - y'}$, $y' = y(t - \Delta t)$

t	x	y'	y
0	100	0	0
t = Δt	100	0	100
t = 2Δt	100	100	0
3Δt	100	0	100

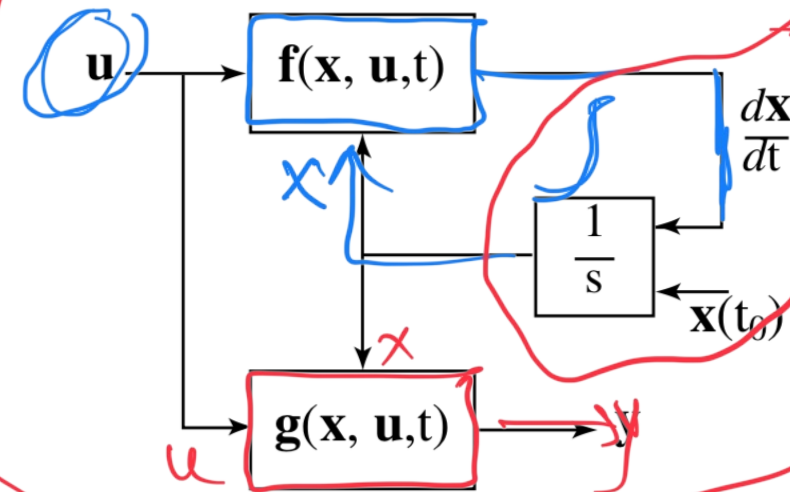
t	x	y'	y
0	100	20	80
Δt	100	80	20
2Δt	100	20	80

State-Variable Solution

- **Objective:** To find a **numerical solution** for the state-space equation below:

General State-Space Form

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x},\ \mathbf{u},\ t) \\ \mathbf{y} = \mathbf{g}(\mathbf{x},\ \mathbf{u},\ t) \end{cases}$$



✓ Assume there is **no algebraic loop**

✓ Assume input $\mathbf{u}$ is part of the definition of the function $\mathbf{f}$ ⇒ treat $\mathbf{u}$ as a **known function of time** and incorporate it inside $\mathbf{f}$

Standard Canonical Form

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{y} = \mathbf{g}(\mathbf{x}, t) \end{cases} \qquad \mathbf{x}(t_0) = \mathbf{x}_0$$

⇒ ✓ The numerical algorithms used to solve this set of ODEs are called **ODE SOLVERS**